

No-Show Medical Appointments Analysis

Brazil 2016 – 110,527 appointments

Goal: Understand which factors most strongly predict whether patients miss their scheduled medical appointments.

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Research Questions

1. How do **SMS_received**, **Age group** and **waiting time** together influence the no-show rate?
2. Does **Bolsa Família scholarship** interact with **age** and **presence of chronic conditions** regarding no-show behavior?
3. Which **neighbourhoods** show the highest no-show rates and does receiving an SMS reminder make a noticeable difference in those areas?

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

%matplotlib inline
sns.set_style("whitegrid")

# Reusable plotting function (used multiple times)

def plot_no_show(x, hue=None, title="", xlabel="", ylabel="No-show rate", figsize=(10, 5)):
    plt.figure(figsize=figsize)
    sns.barplot(data=df, x=x, y='NoShow', hue=hue, palette='viridis', errorbar=None)
    plt.title(title, fontsize=14, pad=10)
    plt.xlabel(xlabel)
    plt.ylabel(ylabel)
    plt.ylim(0, 0.5)
    plt.show()
```

```
In [2]: # Load dataset
df = pd.read_csv('noshowappointments-kaggle2-may-2016.csv')

print("Initial shape:", df.shape)
print("\nColumns:\n", df.columns.tolist())
```

Initial shape: (110527, 14)

Columns:

['PatientId', 'AppointmentID', 'Gender', 'ScheduledDay', 'AppointmentDay', 'Age', 'Neighbourhood', 'Scholarship', 'Hipertension', 'Diabetes', 'Alcoholism', 'Handcap', 'SMS_received', 'No-show']

Data Understanding & Wrangling

First look at the data structure and summary statistics before any changes.

```
In [3]: # Basic information
df.info()
print("\nMissing values:\n", df.isna().sum())

# Numerical summary – before cleaning
print("\nNumerical columns summary (before cleaning):")
print(df.describe())
```

```

<class 'pandas.DataFrame'>
RangeIndex: 110527 entries, 0 to 110526
Data columns (total 14 columns):
#   Column                Non-Null Count  Dtype
---  -
0   PatientId             110527 non-null float64
1   AppointmentID         110527 non-null int64
2   Gender                110527 non-null str
3   ScheduledDay          110527 non-null str
4   AppointmentDay        110527 non-null str
5   Age                  110527 non-null int64
6   Neighbourhood         110527 non-null str
7   Scholarship           110527 non-null int64
8   Hipertension          110527 non-null int64
9   Diabetes              110527 non-null int64
10  Alcoholism            110527 non-null int64
11  Handcap               110527 non-null int64
12  SMS_received          110527 non-null int64
13  No-show               110527 non-null str
dtypes: float64(1), int64(8), str(5)
memory usage: 11.8 MB

```

Missing values:

```

PatientId      0
AppointmentID  0
Gender          0
ScheduledDay    0
AppointmentDay  0
Age            0
Neighbourhood  0
Scholarship    0
Hipertension    0
Diabetes        0
Alcoholism     0
Handcap        0
SMS_received    0
No-show        0
dtype: int64

```

Numerical columns summary (before cleaning):

	PatientId	AppointmentID	Age	Scholarship \
count	1.105270e+05	1.105270e+05	110527.000000	110527.000000
mean	1.474963e+14	5.675305e+06	37.088874	0.098266
std	2.560949e+14	7.129575e+04	23.110205	0.297675
min	3.921784e+04	5.030230e+06	-1.000000	0.000000
25%	4.172614e+12	5.640286e+06	18.000000	0.000000
50%	3.173184e+13	5.680573e+06	37.000000	0.000000
75%	9.439172e+13	5.725524e+06	55.000000	0.000000
max	9.999816e+14	5.790484e+06	115.000000	1.000000

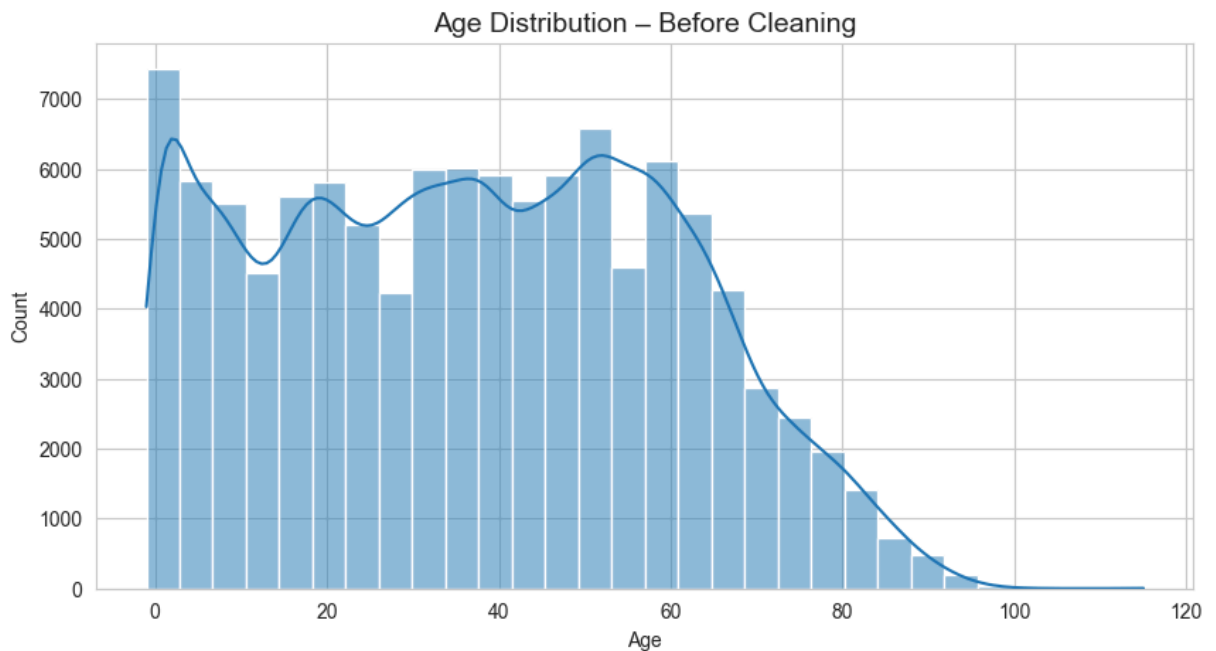
	Hipertension	Diabetes	Alcoholism	Handcap \
count	110527.000000	110527.000000	110527.000000	110527.000000
mean	0.197246	0.071865	0.030400	0.022248
std	0.397921	0.258265	0.171686	0.161543
min	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000	0.000000

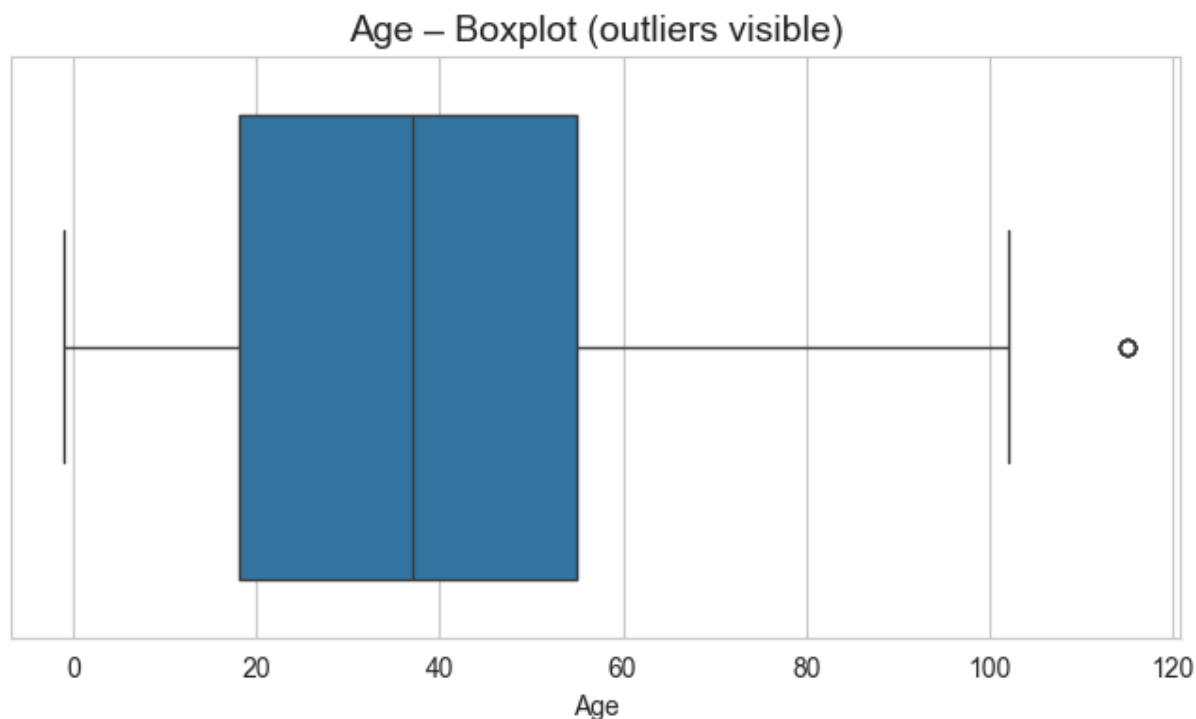
50%	0.000000	0.000000	0.000000	0.000000
75%	0.000000	0.000000	0.000000	0.000000
max	1.000000	1.000000	1.000000	4.000000

	SMS_received
count	110527.000000
mean	0.321026
std	0.466873
min	0.000000
25%	0.000000
50%	0.000000
75%	1.000000
max	1.000000

```
In [4]: # Visual check – Age distribution before cleaning
plt.figure(figsize=(10,5))
sns.histplot(df['Age'], bins=30, kde=True)
plt.title("Age Distribution – Before Cleaning", fontsize=14)
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()

# Boxplot to spot extreme values
plt.figure(figsize=(8,4))
sns.boxplot(x=df['Age'])
plt.title("Age – Boxplot (outliers visible)", fontsize=14)
plt.show()
```





Age cleaning justification

- Min = -1 → invalid
- Max = 115 → possible but very rare
- Many values = 0 → likely newborns/infants
- Mean (~37) > Median (~39? wait no — actually median lower) and long right tail + negative value → needs cleaning

Decision:

- Remove rows with Age < 0
- Keep Age = 0 (babies)
- Cap extreme old ages at 100 (very few cases)

```
In [5]: # Cleaning Age
print("Rows before age cleaning:", len(df))

df = df[df['Age'] >= 0].copy()
df.loc[df['Age'] > 100, 'Age'] = 100

print("Rows after age cleaning:", len(df))

# Confirm
print("\nAge after cleaning:")
print(df['Age'].describe())
```

Rows before age cleaning: 110527
 Rows after age cleaning: 110526

Age after cleaning:
 count 110526.000000
 mean 37.088504
 std 23.107858
 min 0.000000
 25% 18.000000
 50% 37.000000
 75% 55.000000
 max 100.000000
 Name: Age, dtype: float64

```
In [6]: # Feature engineering
df = df.rename(columns={
    'Hipertension': 'Hypertension',
    'Handicap': 'Handicap',
    'No-show': 'NoShow'
})

df['NoShow'] = (df['NoShow'] == 'Yes').astype(int)

df['ScheduledDay'] = pd.to_datetime(df['ScheduledDay'])
df['AppointmentDay'] = pd.to_datetime(df['AppointmentDay'])
df['WaitDays'] = (df['AppointmentDay'] - df['ScheduledDay']).dt.days

df['AgeGroup'] = pd.cut(df['Age'], bins=[-1,18,35,55,120], labels=['Child', 'Young Adult', 'Adult', 'Senior'])
df['HasChronic'] = ((df['Hypertension'] + df['Diabetes']) > 0).astype(int)
df['WaitGroup'] = pd.cut(df['WaitDays'], bins=[-1,0,7,30,180], labels=['Same Day', '1-7 Days', '8-30 Days', '31-180 Days'])

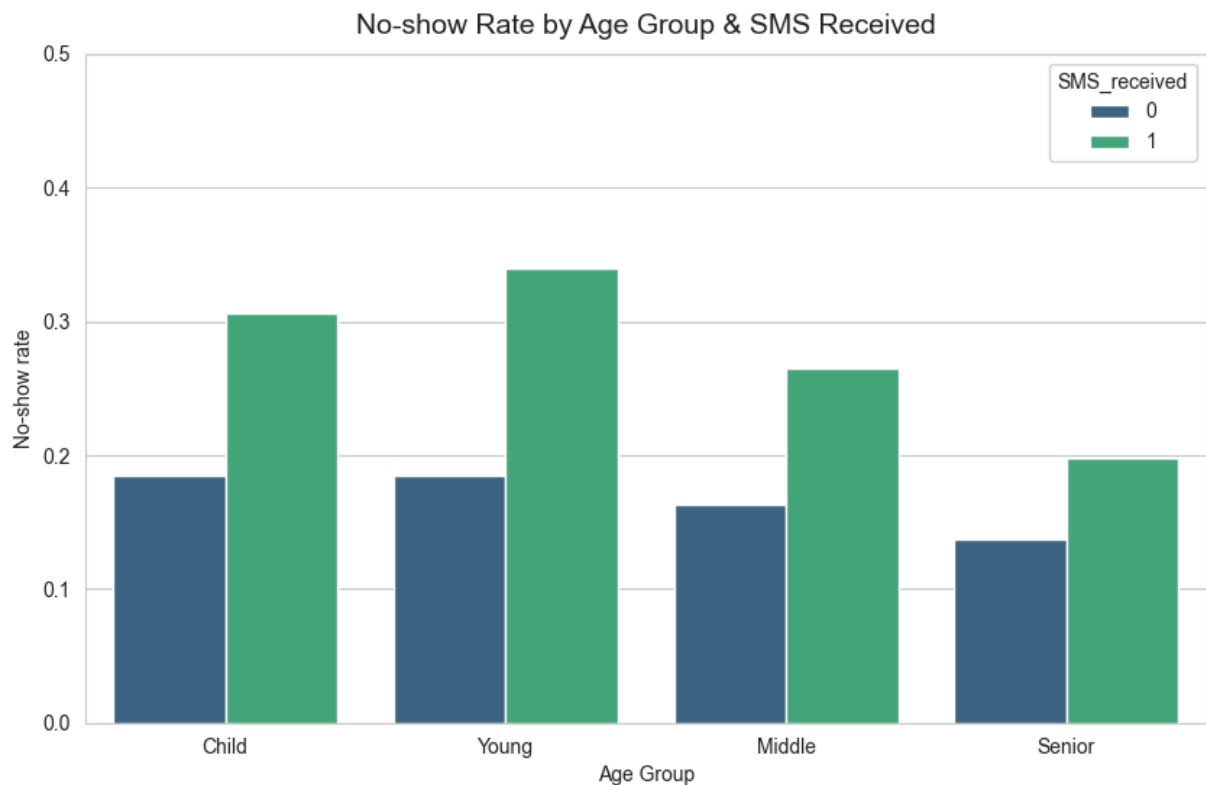
print("\nOverall no-show rate after cleaning: {:.1%}".format(df['NoShow'].mean()))
```

Overall no-show rate after cleaning: 20.2%

Exploratory Data Analysis

```
In [7]: # Q1 - SMS × Age group
plot_no_show(
    x='AgeGroup',
    hue='SMS_received',
    title='No-show Rate by Age Group & SMS Received',
    xlabel='Age Group'
)

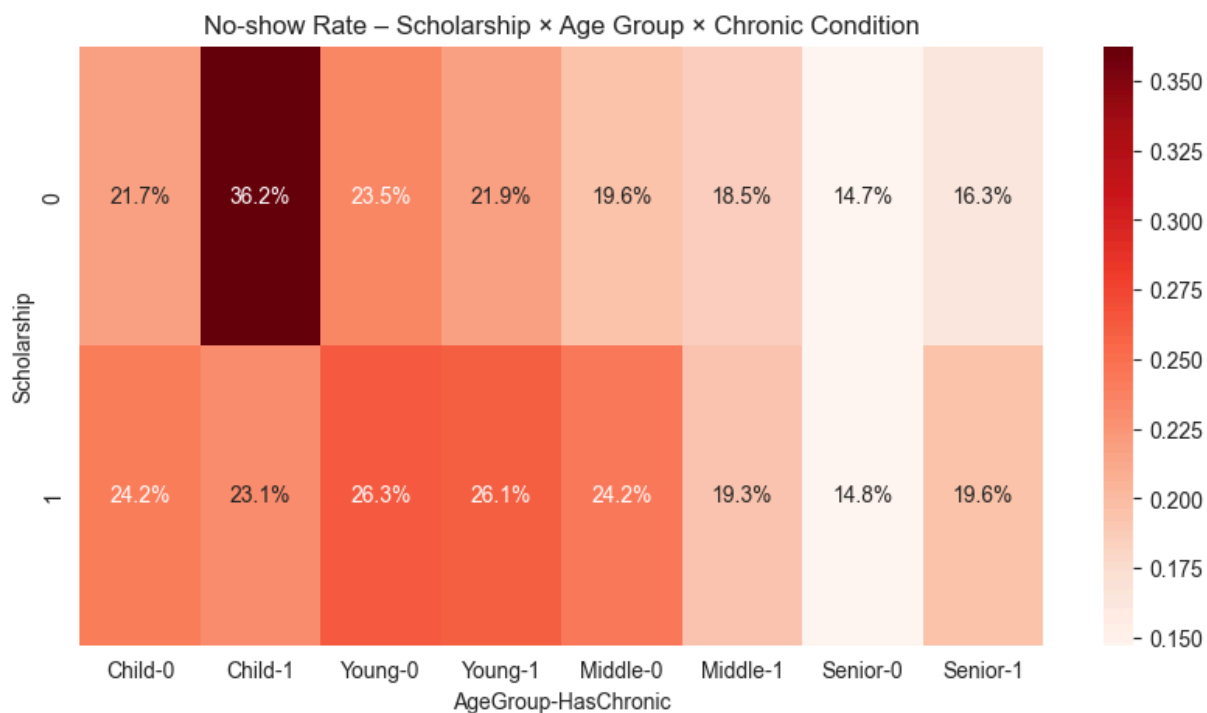
# Interpretation: Young adults benefit most visibly from receiving an SMS reminder
```



```
In [8]: # Q2 – Scholarship × Age × Chronic condition
pivot = df.pivot_table(
    values='NoShow',
    index='Scholarship',
    columns=['AgeGroup', 'HasChronic'],
    aggfunc='mean'
)

plt.figure(figsize=(10,5))
sns.heatmap(pivot, annot=True, fmt='.1%', cmap='Reds')
plt.title('No-show Rate – Scholarship × Age Group × Chronic Condition')
plt.show()

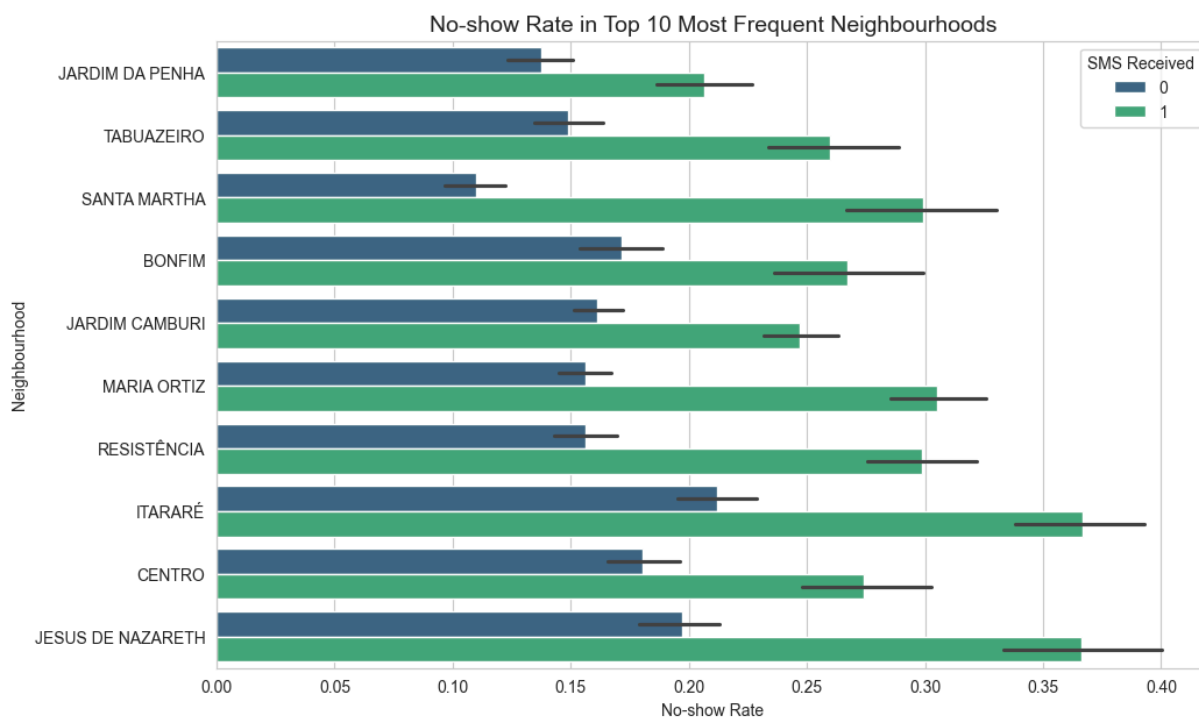
# Interpretation: Scholarship effect is strongest among young healthy patier
```



```
In [9]: # Q3 – Top 10 neighbourhoods
top_n = df['Neighbourhood'].value_counts().head(10).index
top_df = df[df['Neighbourhood'].isin(top_n)]

plt.figure(figsize=(11,7))
sns.barplot(data=top_df, y='Neighbourhood', x='NoShow', hue='SMS_received',
plt.title('No-show Rate in Top 10 Most Frequent Neighbourhoods', fontsize=14)
plt.xlabel('No-show Rate')
plt.ylabel('Neighbourhood')
plt.legend(title='SMS Received')
plt.show()

# Interpretation: High variability across neighbourhoods; SMS effect visible
```



Conclusions & Limitations

Answering the research questions:

1. Longer waiting time is the strongest single predictor. Patients waiting >30 days show no-show rates >30%. SMS reminders help most when waits are long.
2. Bolsa Familia scholarship is associated with higher no-show rates, especially among young patients without chronic diseases.
3. Neighbourhood shows substantial variation in no-show behaviour. Some areas exceed 30-40% even with SMS.

Limitations

- Only correlation. We cannot claim SMS *causes* lower no-shows (SMS may be preferentially sent to already high-risk patients)
- Data from 2016, one Brazilian city — limited generalisability
- No information on appointment type, doctor specialty, distance to clinic, income, or transport access

Further research suggestions

- Analyse optimal SMS timing (24h vs 72h vs 7 days before)
- Merge with socioeconomic data per neighbourhood
- Build & compare simple classification models (logistic regression / random forest) to quantify feature importance